FIN 3080 Tutorial 2

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Sth about tutorials

Orientation

- Provide detailed instructions for homework implementations
- Usually self contained (for homework)

Structure

- Empirical intuitions and designs
- Stata functions and their usages
- Real screen outputs from code implementation (whenever possible)

Sth about homework

Basic elements

- Emprical analysis on the Chinese stock market
- Any programming languages are good but materials are mostly in Stata
- General academic integrity disciplines apply

Submission requirements

- A pdf document containing arguments, results (tables, figures, etc), discussions; 11pt, 1.5 spaced, $\leq$ 5 pages
- A zipped package of code scripts that generate the results

Suggested solution (post-submission)

- A zipped package of *data*, *codes*, and *results*; uploaded after DDL
- LLMs may help translate Stata codes into Python/R/... codes



Sth about TAs

Full-time TAs $\times$ 2:

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For efficient Q&As:

- Check the correct e-mail addresses and office hours
- Provide scripts/codes, data, and screen outputs; ~~WeChat~~, ~~小作文~~



General methodology

Usually it takes five steps to finish homework/conduct empirical analysis

Step 1. Download data from CSMAR

Step 2. Process raw data

Step 3. Merge data sets

Step 4. Analyze data

Step 5. Report & discuss results

Step 1. Download data from CSMAR

We can access CSMAR (https://data.csmar.com/) with our university IPs.

When off the campus, we may use the university VPN (see https://itso.cuhk.edu.cn/en/node/169)

Key elements in data downloading:

- variable name, definition, and unit
- data coverage (usually company & time period)
- data frequency

Step 1. Download data from CSMAR

Let's dive into Problem 1 from Homework 1:

Problem 1. *Please access Stock Trading, Financial Statements and Financial Indicators databases on CSMAR and download the following data for all listed firms in the A-share market: (i) monthly stock prices, stock returns, market value of tradable shares from Jan. 2000 to Sep. 2023; (ii) quarterly total assets, total liabilities, earnings per share, ROA (return on asset), ROE (return on equity), R&D expenses from 2000Q1 to 2023Q3; (iii) establishment date and market type. Then complete the following two tasks based on the downloaded data.*

To get familiar with the database quickly, try to play with CSMAR with the assistance its official documentations and refer to the 1st tutorial.



Step 2. Process data

Data processing usually includes following procedures:

2.1 Load data

- load .csv files with *insheet*; load .dta files with *use*
- Stata only allows us to process one data set a time
    * when load data, we need to specify the *clear* option
    * when finish processing data, we usually *save* data locally

Step 2. Process data

2.2 Check and format variable types

- The most tricky and important variables are **dates**
  * dates are usually read as strings when loaded in Stata
  * Stata has many built-in types of dates, e.g., *year-quarter, year-month, year-month-day* and functions to do the convertion
  * built-in date types can be converted interchangeably: $YMD \Rightarrow YM \Rightarrow YQ$
  * built-in date types are equipped with many powerful implementations, e.g., *leading, lagging* and etc.

An example of converting string "YYYY-MM" dates into Stata's YM-type dates:

```
gen date_ym = monthly(raw_date, "YM") // e.g., "2019-11" to "2019m11"
    format date_ym %tm // to make dates human-friendly
```

Step 2. Process data

2.3 Construct new variables of interest

- we may use *rename* to change variable names
- we may use *gen* to create new variables
- we may use *replace* to change variable values

2.4 Refine sample coverage

- we may use *drop* to exclude variables or samples

2.5 By-group and conditioning processing

- almost all processing commands in Stata can be combined with *if* and *bysort*



Step 2. Process data

In so doing, we may arrive at the monthly price data:

stock_code	date_ym	...	price
1	2010m1	...	12.8
1	2010m2	...	16.02
⋮	⋮	...	⋮
10086	2020m10	...	123
10086	2020m11	...	120.12
10086	2020m12	...	118.7

Step 2. Process data

And quarterly earnings per share data:

stock_code	date_yq	...	eps
1	2010q1	...	2
1	2010q2	...	2.62
⋮	⋮	...	⋮
10086	2020q2	...	0.7
10086	2020q3	...	0.92
10086	2020q4	...	1.43

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Step 3. Merge data

To conduct financial empirical analysis, we unavoidably need to merge different data sets, usually one from secondary market trading and one from financial statements.

Key elements in merging:

- merging key(s):
  - Stata: keys must be the same across data sets
  - Python: keys may have different names across data sets
- merging type:
  - Stata: 1 to 1, 1 to many, many to 1, many to many ...
  - Python: inner, outer, left, right ...

Step 3. Merge data

Take Problem 1.(b) as an example, which requires to construct monthly *P/E ratios*

Assume we have finished processing monthly stock *closing prices* and quarterly *earnings per share*.

But how to estimate it with real data - noting that the frequencies for price and eps are not consistent?

- The common strategy is to match with the lastest eps information

For example, we may construct the *P/E ratio* for company i at 2019m11 (i.e., Nov. 2019) as follows:

$$P/E_{\,i,2019m11} = \frac{Closing\,price_{i,2019m11}}{Quarterly\,earnings\,per\,share_{i,2019q3}}.$$

To make our lifes easier, try to use *EPS-TTM* from CSMAR.



Step 3. Merge data

Empirically we need to construct a new column for **lagged** quarterly dates in the monthly stock price data - *date_yq*

stock_code	date_ym	...	price	date_yq
1	2010m1	...	12.8	2009q4
1	2010m2	...	16.02	2009q4
⋮	⋮	...	⋮	⋮
10086	2020m10	...	123	2020q3
10086	2020m11	...	120.12	2020q3
10086	2020m12	...	118.7	2020q3



Step 3. Merge data

Then merge the two data sets on **stock_code** and **date_yq**

stock_code	date_ym	...	price	date_yq
1	2010m1	...	12.8	2009q4
1	2010m2	...	16.02	2009q4
⋮	⋮	...	⋮	⋮
10086	2020m10	...	123	2020q3
10086	2020m11	...	120.12	2020q3
10086	2020m12	...	118.7	2020q3

Merge into

stock_code	date_yq	eps
1	2010q1	2
1	2010q2	2.62
⋮	⋮	⋮
10086	2020q2	0.7
10086	2020q3	0.92
10086	2020q4	1.43



Step 3. Merge data

In Python: *pandas dataframes* have built-in methods *merge*

In Stata: the key function is *merge*

- Usage: *merge m:1 date_ym using processed_market_rt*

Example for this case:

```
use processed_stock_data, clear
    merge m:1 stock_code date_yq using processed_eps_data
```

Remark: after calling *merge*, a new variable named '*_merge*' will be created. Try to figure out how this variable matters.



Step 3. Merge data

Then we should obtain sth like this:

stock_code	date_ym	price	...	date_yq	eps
1	2012m11	12.8	...	2012q3	2
1	2012m12	16.02	...	2012q3	2
1	2013m01	21.1	...	2012q4	2.62
1	2013m02	23	...	2012q4	2.62
1	2013m03	20.12	...	2012q4	2.62
1	2013m04	18.7	...	2013q1	0.92
1	2013m05	20.99	...	2013q1	0.92

Then it will be straightforward to derve *P/E ratios*.

Remark: the same trick apply to *P/B ratios* as well



Step 4. Analysize data

Homework 1 problem 1 only involves providing summary statistics by market type.

Summarizing statistics can be accomplished by *summarize* in Stata

- Usage: *summ v1 v2 ... vn [if] [, options]*
- Common option is *detail*, which provide additional statistics

We can summarize varibles by group (i.e., market type here) with *if*

```
summ var1 var2 var3 if market_type == "Main", detail
```

Or with *bysort*

```
bysort market_type: summarize var1 var2 var3, detail
```

In Python: *pandas*'s built-in *apply()*, *get_dummies()* and *describe()* may help



Step 5. Report & discuss results

We should arrive at sth like this:

Variable	Obs	Mean	STD	25%	Median	75%
Age	41,711	27.11	7.88	21	25	31
Gender	41,711	0.18	0.39	0	0	0
Score	41,711	56.76	15.26	48.00	60.20	66.75
Delinquency	97,783	0.43	0.50	0	0	1
Approval	236,967	0.41	0.49	0	0	1

Table 1: Example from Dai, Han, Shi, and Zhang (2020)

- Missing/NaN values should be excluded from the table

Most intuitively, paste the table or take a screenshot and attach it in the report.

But there are more elegant ways to export results from Stata.



Export summary statistics elegantly (For your own interest only)

Install with *ssc*

```
ssc install estout
```

Post summary statistics results with estpost

```
estpost summarize stock_ret market_value firm_age total_asset \\\
rd_expense total_profit pe_ratio pb_ratio if market_type == "Main", detail
```

Store posted results with *eststo*

```
eststo ss_main
```

Then export statistics with esttab

```
esttab ss_main, cell(b(fmt(2)) "count mean p25 p50 p75 sd" ), \\\
using ss_main.csv, replace
```

Try to figure out the meaning of these options by yourself!

Tips

Tips for homework

- Try to touch data early instead of catching DDLs
- Order thoughts before diving into coding; aviod unnecessary loops
- Seek help from TAs, Google, documentations ...
- Writing and typesetting are not graded but they do matter in industry and academy

Functions used in this tutorial

- gen, egen, rename, replace, dofm/qofd/monthly, xtset, summarize, merge, format, bysort, duplicates, ssc, eststo, esttab

Special notes on dates in Stata

- Working with dates and times in Stata

